TIPS

Devices/Technology	Alternative Names
Ethernet Technology	RJ45, 802.3 (according to IEEE)
RJ45 Connector (male)	RJ45 plug, Ethernet connector, 8P8C connector
RJ45 socket (female)	Rj45 jack, Ethernet Port
RJ45 Cable	Ethernet cable,

13.3.4 Crimping Tool

Crimping is the process of joining two or more pieces of metal or wire by deforming one or both of them to hold each other. Joining RJ45 connector together with twisted pair cable at each end is an essential process in Ethernet cabling which lead the cable to function properly.

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The crimping tool is a physical tool which is used to connect the patch wire and the Ethernet connector. The crimping tool looks like a small cutting handle with two mould of Ethernet port. The tool will puncture the connector and inserts the wire set in the connector. See Figure 13.14

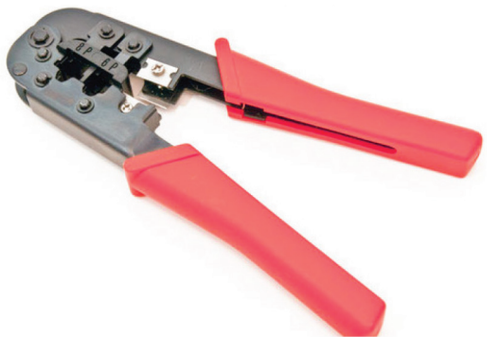


Figure 13.14 Crimping Tool for RJ-11 (6 pin) and RJ-45 (13 pin)

Crimping process for making Ethernet cables

1. Cut the cable with desired length
2. Strip the insulation sheath about 1 inch from both end of the cable and expose the Twisted pair wires
3. After stripping the wire, untwist the smaller wires and arrange them into the proper wiring scheme, T568B preferred generally.
4. Bring the wires tighter together and cut them down so that they all have the same length (½ inch).
5. Insert all the 8 coloured wires into the eight grooves in the connector. The wires should be inserted until the plastic sheath is also inside the connector.
6. Use the crimping tool to lock the RJ45 connector on the cable. It should be strong enough to handle manual traction. Now it is ready for data transmission.
7. Use a cable tester to verify the proper connectivity of the cable. See Figure 13.15

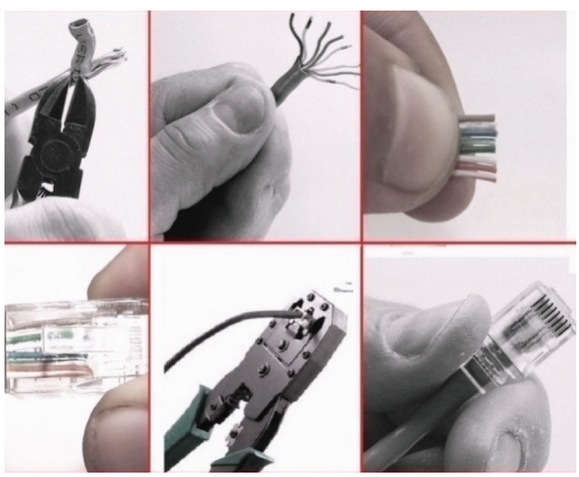


Figure 13.15 Crimping process using crimping tool

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